

Standard Operating Procedure ***TIG 275 Welder***

Lincoln Electric Precision TIG 275

The **Precision TIG 275** is a versatile power source delivering an output range of **2 to 340 amps**. It features patented **Micro-Start™ II** technology for enhanced arc stability and **AC Auto Balance** for optimized cleaning and penetration on aluminum. This machine is a high-precision instrument intended for technical maintenance and fabrication applications.

Potential Hazards

- **Electric Shock**: The electrode and work circuits are electrically “hot” when the welder is on; touching these parts with bare skin or wet clothing can be fatal.
- **Arc Rays**: Ultraviolet and infrared rays can cause severe burns to eyes and skin; nearby personnel must be protected by non-flammable screening.
- **Fumes and Gases**: Welding on coated or plated parts (e.g., galvanized steel) can release highly toxic gases like phosgene if chlorinated solvents are present.
- **Cylinder Explosion**: Compressed gas cylinders must be securely chained in an upright position; a damaged valve can turn a cylinder into a projectile.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Welding Helmet	**ANSI Z87.1 Compliant:** Must use a properly fitted helmet with the correct grade of filter plate (Shade 10-13 depending on amperage) to protect against arc rays.
Eye Protection	Wear ANSI Z87.1-compliant safety goggles. Mandatory under the helmet to protect against slag and debris when the helmet is raised.
Hand Protection	**Dry, Hole-Free Welding Gloves:** Leather TIG gloves are required to insulate hands from heat and "hot" electrical circuits.
Body Protection	**Flame-Resistant Apparel:** Wool or heavy cotton clothing, leather aprons, and high boots are required to prevent burns from arc flash and spatter.
Respiratory	**Fume Extraction: Use local exhaust ventilation to keep the breathing zone clear of welding fumes.**

Before Use

- **Cable Routing**: Route the electrode and work cables together and secure them; never coil the electrode lead around your body.



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- **Connection Check**: Ensure the work clamp makes a solid electrical connection to the workpiece as close as possible to the weld area.
- **Gas Setup**: Verify the shielding gas (typically 100% Argon for TIG) is securely connected and set the flow meter between **10 and 25 CFH**.
- **Torch Inspection**: Check the TIG torch for worn collets or cracked alumina nozzles; ensure the tungsten electrode is properly ground to a point for DC or a balled end for AC.

Operational Procedures

- **Starting the Arc**: Stand to the side of the machine during power-up. Use the foot pedal to initiate the arc without touching the tungsten to the workpiece (HF Start).
- **Heat Management**: Maintain a consistent arc length (approx. 1/16 to 1/8 inch). Use the foot pedal to modulate amperage in real-time based on the puddle's behavior.
- **Termination**: Use the **Post-Flow** timer (set to 10-15 seconds) to shield the cooling weld and tungsten from atmospheric contamination after the arc is extinguished.

During Use

- **Arc Length & Torch Control:** Maintain a consistent, tight arc length between 1/16" and 1/8" off the work surface. Maintain a 70° to 80° push travel angle with a 90° work angle to provide uniform shielding gas coverage over the molten puddle.
- **Tungsten Contamination:** If the tungsten electrode touches the weld puddle or filler rod, extinguish the arc immediately. Never continue welding with contaminated tungsten—shut off the pedal, remove the electrode, and re-grind it before resuming.
- **Filler Rod Technique:** Feed filler rod into the leading edge of the puddle within the inert gas envelope. Keep the hot end of the filler wire inside the argon shield at all times to prevent atmospheric oxidation and joint contamination.
- **Heat Modulation:** Modulate the foot pedal continuously to maintain puddle control. Taper off amperage as base metal heat saturation increases to prevent keyholing, burn-through, or excessive heat-affected zone (HAZ) distortion.
- **Crater Fill & Post-Flow Dwell:** Ramp down pedal pressure slowly at the end of the bead to fill the crater and prevent solidification cracking. Hold the torch stationary over the cooling weld pool for the full post-flow cycle (10–15 seconds) until the argon stops flowing and the tungsten cools.



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- **Electrical Isolation:** Keep your body insulated from the workpiece and weld table. Never lean on the table or rest bare arms on grounded metal while the machine is energized or while initiating the arc with high frequency (HF).
- **Thermal Handling:** Welded assemblies retain extreme temperatures long after arc extinguishment. Always manipulate hot parts with locking pliers or tongs; never pick up freshly welded stock with bare gloves.
- **Draft & Shielding Monitoring:** Check for drafts from open shop doors, fans, or fume extractors that could disrupt the argon envelope. If porosity, black soot, or wandering arcs occur, stop immediately and check gas flow and nozzle condition.

After Use

- **System Shutdown**: Turn off the power switch and close the gas cylinder valve.
- **Line Purge**: Depress the foot pedal briefly to purge remaining gas from the lines before storage.
- **Housekeeping**: Coil cables neatly on the side-mounted torch holders. Sweep the area of metal off-cuts and debris.
- **Hygiene**: Wash hands thoroughly to remove metallic oxides and particulates after handling filler rods and workpieces.

Emergency Equipment and Procedures

- **Emergency Stop**: Immediately release the foot pedal or trigger and toggle the main power switch to the **OFF** position.
- **Thermal Overload**: If the **Thermal Shutdown Light** illuminates, allow the fan to run with the machine idle until it cools; do not shut off power immediately as this stops the cooling fan.
- **Reporting**: Report all electrical shocks, near-misses, burns, or equipment damage to the Laboratory Project Supervisor.